# Type MS · Describe genetic modification and gene therapy for human health

## Answer 1 of 4 · Question 6 part whole · 6 marks

2022 JUN · Paper 01 · find part whole in this mark scheme extract

| | | | |
|---|---|---|---|
| 6 | Any six from the following:<br><br>• remove plasmid from bacteria;<br>• locate the human insulin gene<br>• cut/remove insulin gene (from human);<br>• cut open the plasmid;<br>• using restriction enzymes;<br>• add the human insulin gene to the plasmid;<br>• use of ligase;<br>• ref recombinant DNA;<br>• return the plasmid to a bacterial cell; | Suitably drawn **and** annotated diagrams can score full marks. | 6 |
| | | | Total 6 |

| Question number | Answer | Notes | Marks |
|---|---|---|---|

# Type MS · Describe genetic modification and gene therapy for human health

## Answer 2 of 4 · Question 6 part c · 3 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 6 | (a) | (i) | Three from:<br>• the use of embryonic stem cells is not needed/embryos not destroyed;<br>• easily available source of cells/easy to obtain/no need to find a donor;<br>• same cell/tissue/blood type of the patient/same antigens;<br>• reduced chance of rejection/not recognised as foreign; | | 3 |
| | | (ii) | Two from:<br>• unlicensed clinics could be used;<br>• stem cells could develop into tumours/cancer;<br>• risk of rejection (if the cells are not the recipients);<br>• increased risk of infection; | | 2 |
| | (b) | (i) | 860 000 and 480 000;<br>380 000; | Ecf allow one mark for incorrectly derived figures from graph but correct answer from these figures | 2 |
| | | (ii) | **D** structure Z; | | 1 |
| | (c) | | Three from:<br>• cut out desired/beta-carotene/gene/gene for vitamin A;<br>• using/cut with restriction enzymes;<br>• insert into plasmid;<br>• insert into/infect rice genome;<br>• to produce golden rice; | | 3 |

Total 11 marks

## Type MS · Describe genetic modification and gene therapy for human health

### Answer 2 of 4 · Question 6 part c · 3 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Describe genetic modification and gene therapy for human health

### Answer 3 of 4 · Question 7 part e · 2 marks

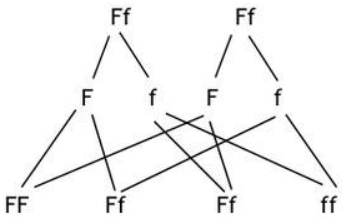
2025 JUN · Paper 02 (regional R) · find part e in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 7 | (a) | amylase; | | 1 |
| | (b) | Any two from<br>• enzymes are immobilised on testing strip;<br>• react with/breaks down glucose (in urine);<br>• (catalyse a chemical reaction that) causes a colour change (in strip); | | 2 |
| | (c) | pancreas; | | 1 |
| | (d) | • rejection/immune response/allergies;<br>• (immune system) recognised it as foreign/antigen; | | 2 |
| | (e) | any two from<br>• use restriction enzymes;<br>• cut out insulin gene from human DNA;<br>• insert human insulin gene into yeast genome/plasmid/genetic material; | | 2 |
| | | | **Total for Question 7 = 8 marks** | |

Type MS · Describe genetic modification and gene therapy for human health

Answer 4 of 4 · Question 8 part b · 3 marks

2024 JUN · Paper 02 · find part b in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 8 | (a) (i) | **B** individual 3<br><br>The following are incorrect because:<br><br>**A** individual 1 is heterozygous<br>**C** Individual 5 is heterozygous<br>**D** individual 3 is homozygous dominant | | (1) |
| | (ii) | **D** 4<br><br>The following are incorrect because:<br><br>**A** there are more than 1 heterozygous genotypes<br>**B** there are more than 2 heterozygous genotypes<br>**C** there are more than 3 heterozygous genotypes | | (1) |
| | (iii) | • they have a dominant allele;<br>• over-rides/ masks the effect of the recessive allele/ must have two recessive alleles for the condition to show/ must have homozygous recessive alleles ; | do not allow dominant allele overpowering the recessive allele | (2) |
| | (iv) | • heterozygous parent genotypes;<br>• gametes;<br>• genotypes of offspring;<br>• offspring genotypes linked to the phenotype for the disease and showing that showing that 1 of the 4 genotype has the disease;<br><br>  <br> | mps 2,3 and 4 marks for correct Punnett square showing all details | (4) |
| | (b) | • reference to use of restriction enzyme<br>• (to obtain) the normal/ non faulty/ dominant allele);<br><br>• use of virus/ correctly named virus;<br>• inserts allele into person **cells**;<br>• functional/normal protein produced; | Accept liposomes ignore plasmids<br><br>Penalise one for using genes instead of alleles. | (3) |

**Total for Question 8 = 11 marks**